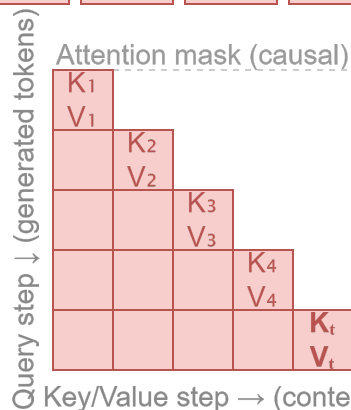
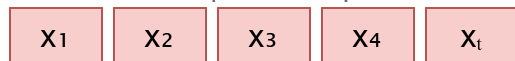


Without KV Cache — Repeated Forward

Generation step t : recompute all tokens



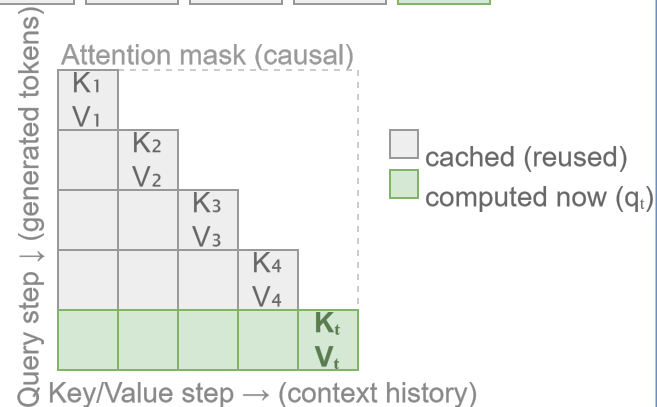
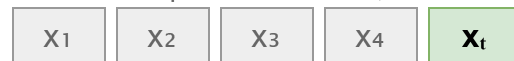
All cells recomputed every step:
self-attention over all N tokens

Per-step cost $\approx O(N^2)$

No reuse: K, V for $x_1 \dots x_t$ are
recomputed at every step

With KV Cache — State Reuse

Generation step t : reuse cache, add one token



Only the new row (q_t) is computed;
previous K, V are reused from the cache

Per-step cost $\approx O(N)$

Past Key/Value Cache (VRAM)
 $K_1, V_1 \dots K_{t-1}, V_{t-1}$ (cached)

Compute Q_t, K_t, V_t
(new token only)

append K_t, V_t

KV Cache reuses cached Key/Value states to cut the per-step decoding cost from $O(N^2)$ to $O(N)$; it does not remove the $O(M^2)$ cost of the initial prompt prefill.